COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

Full Stack Web Development

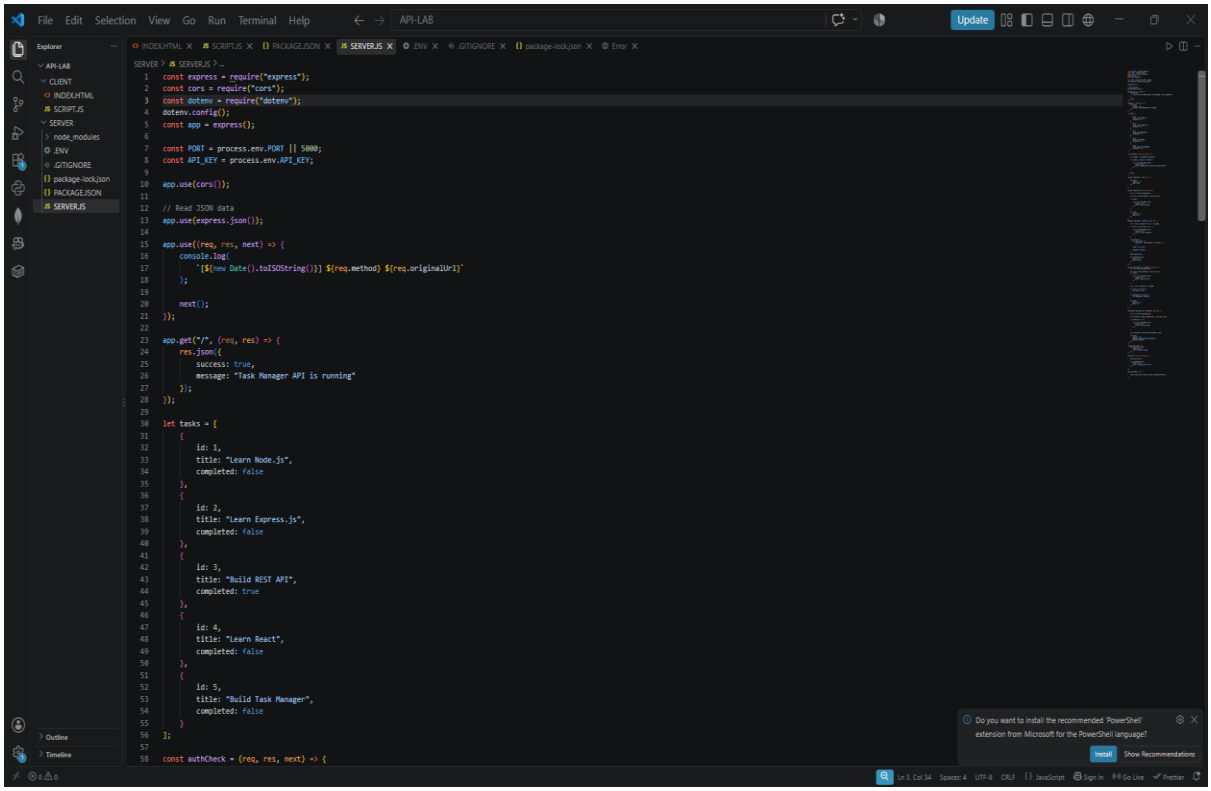
Lab Sheet 6— APIs (Practical)

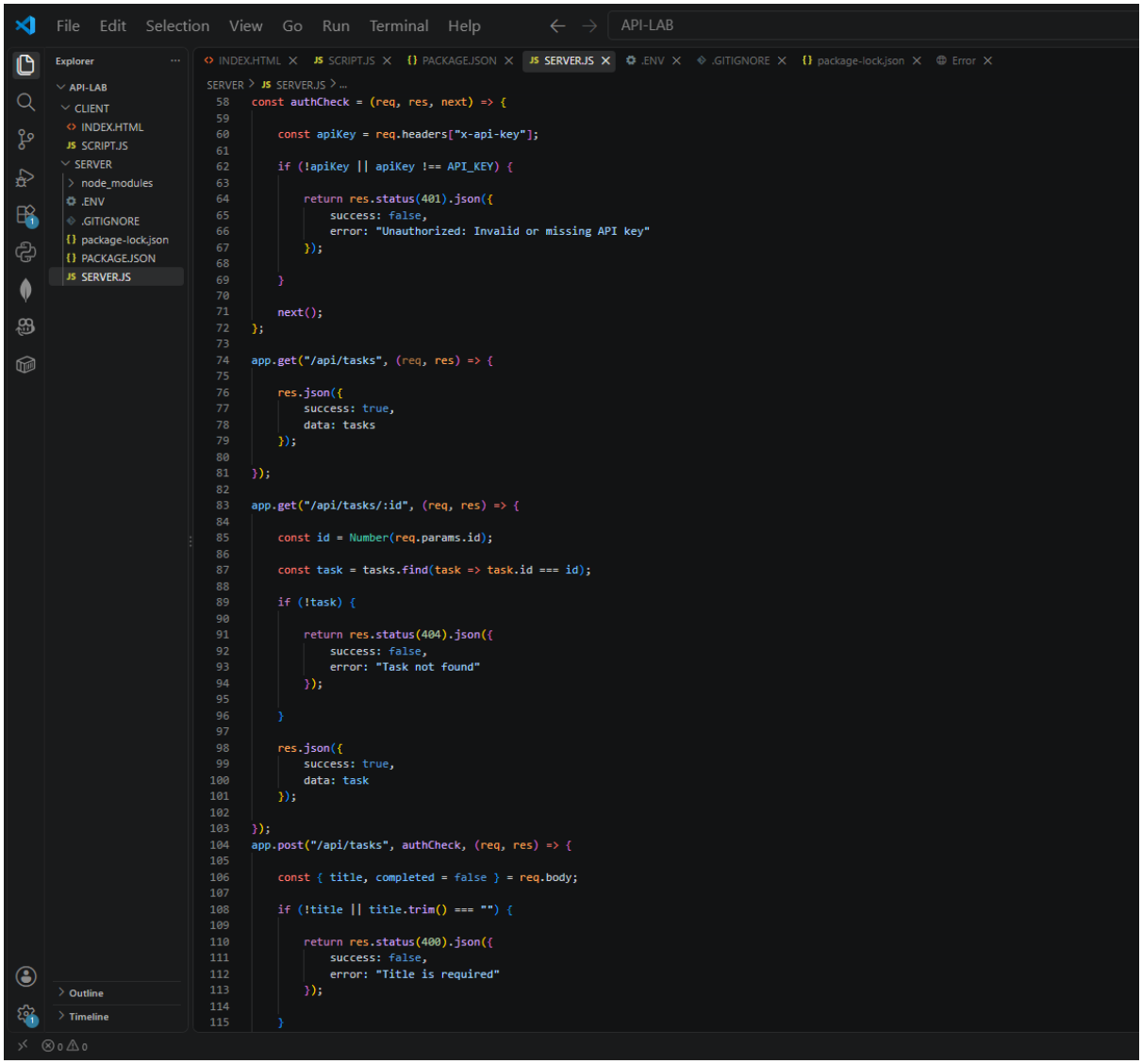
Full Backend Integration: Node.js + Express.js + REST APIs (No Database)

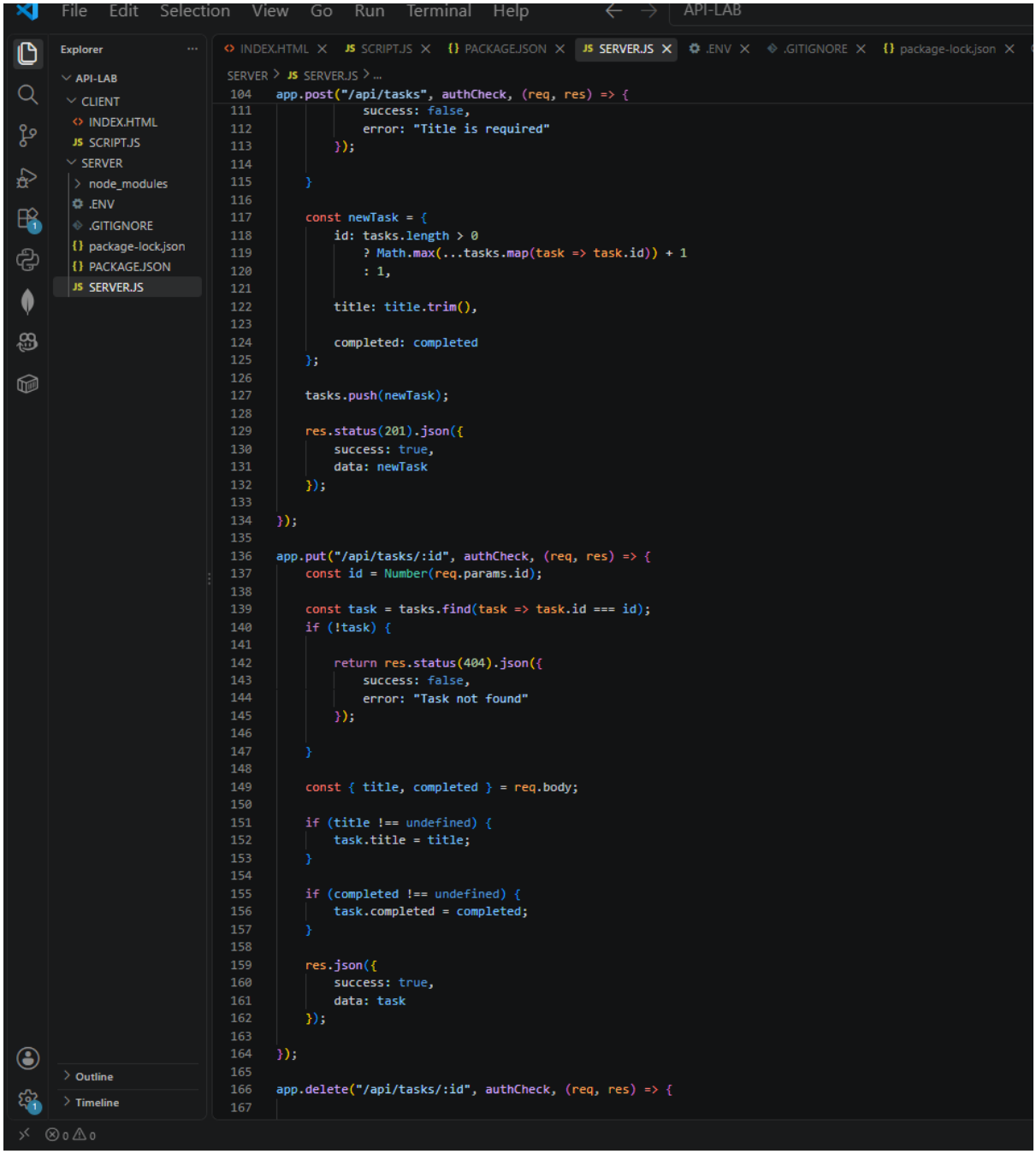
NAME – NITIN DIXIT

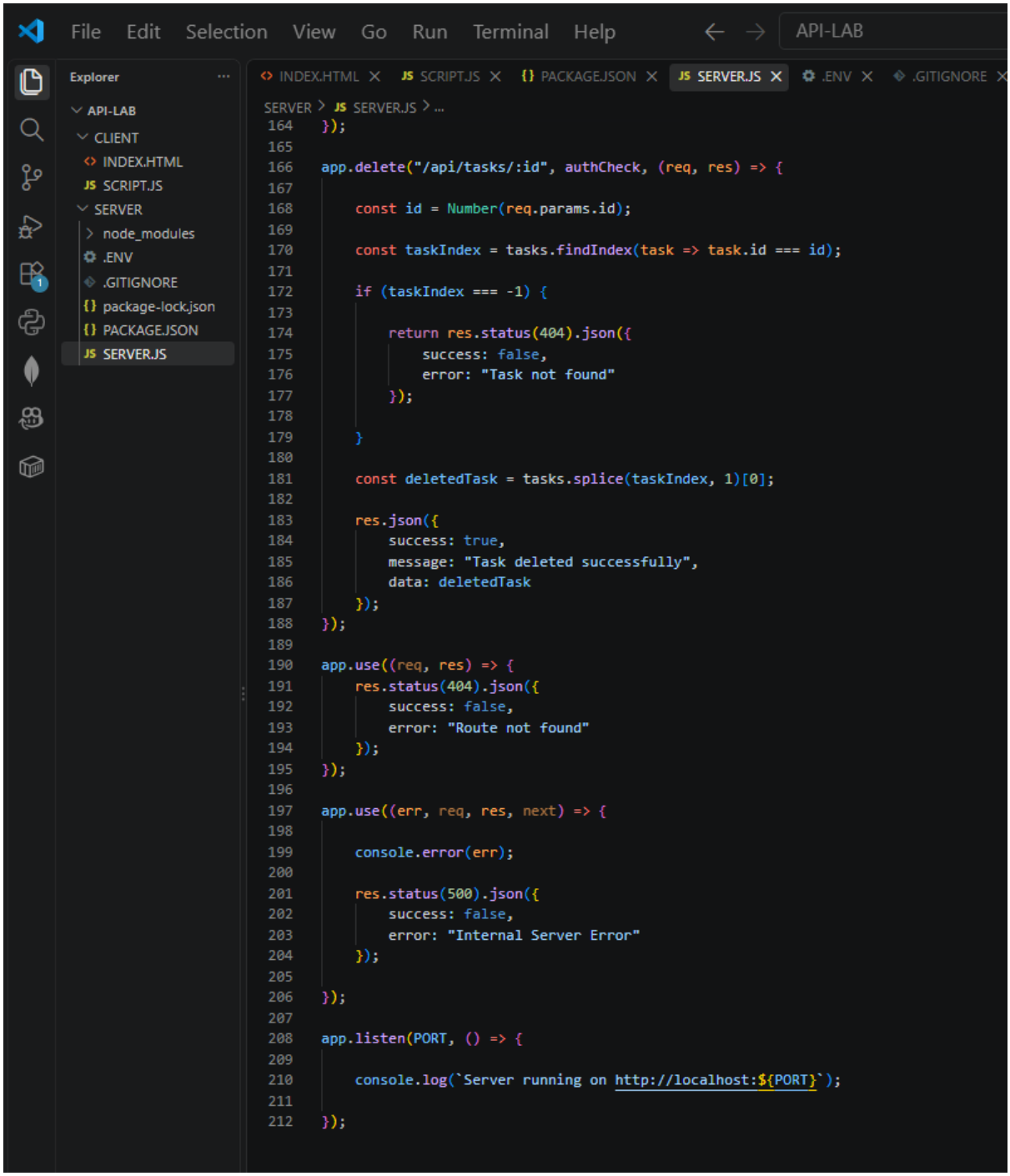
SECTION – B

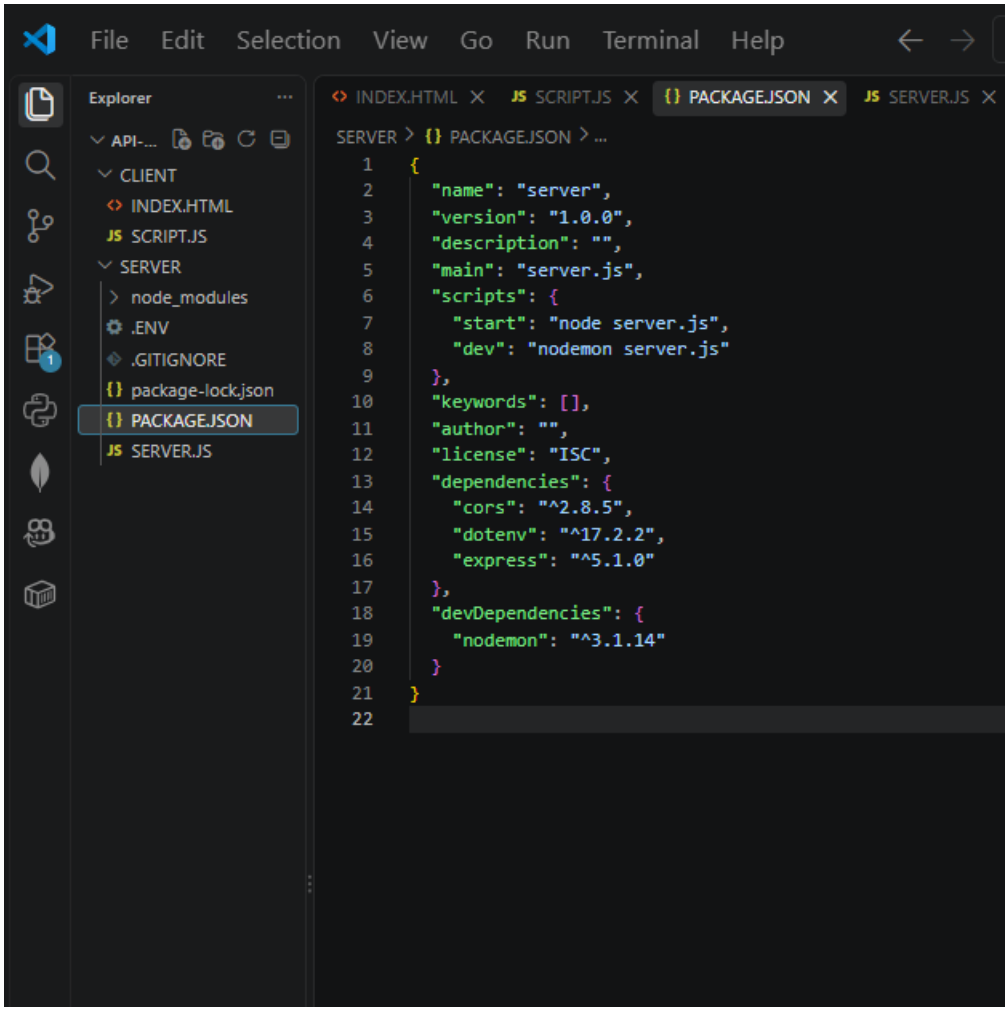
ROLL NUMBER – 38

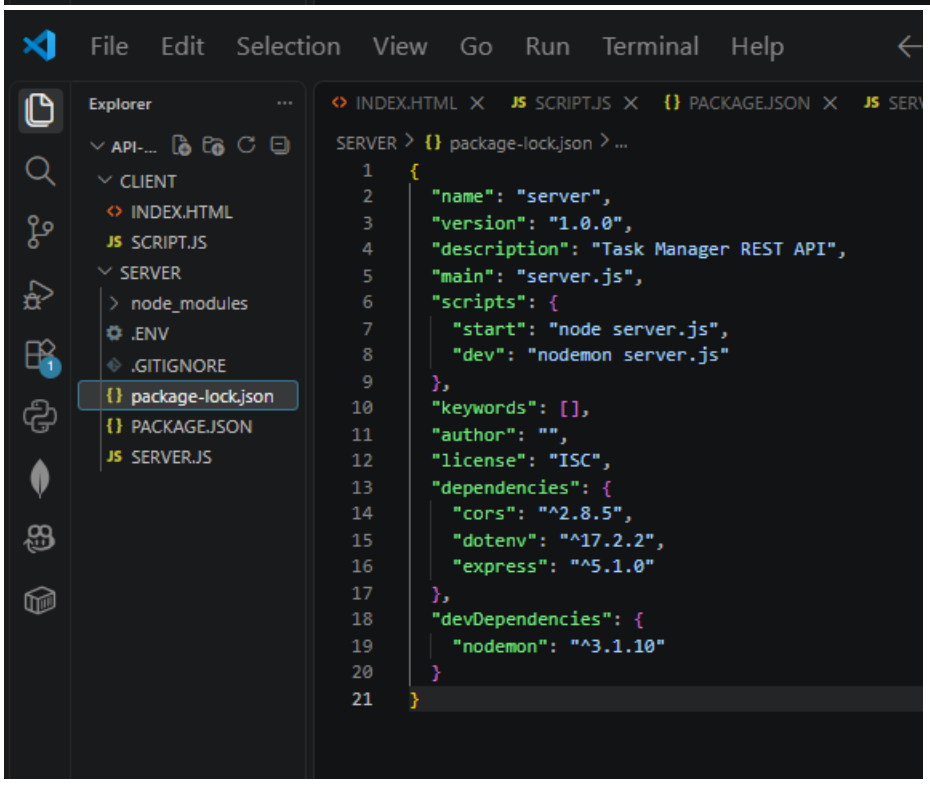


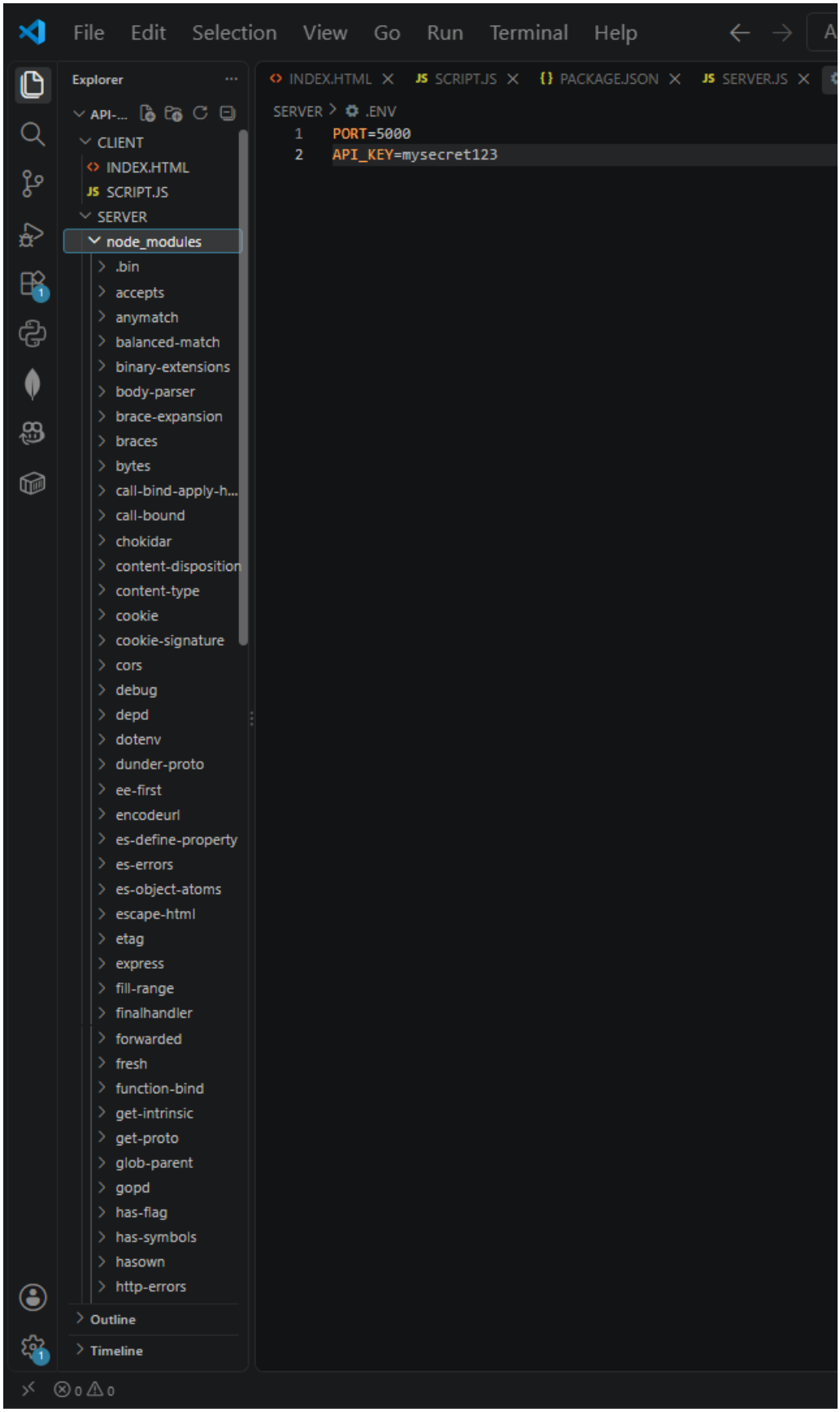

File Edit Selection View Go Run Terminal Help
API-LAB
Explorer
API-LAB
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.ENV
.GITIGNORE
package-lock.json
PACKAGE.JSON
SERVER.JS
Outline
Timeline
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
.ENV
.GITIGNORE
package-lock.json
Error
SERVER > SERVER.JS > ...
const authCheck = (req, res, next) => {
const apiKey = req.headers["x-api-key"];
if (!apiKey || apiKey !== API_KEY) {
return res.status(401).json({
success: false,
error: "Unauthorized: Invalid or missing API key"
});
}
next();
};
app.get("/api/tasks", (req, res) => {
res.json({
success: true,
data: tasks
});
});
app.get("/api/tasks/:id", (req, res) => {
const id = Number(req.params.id);
const task = tasks.find(task => task.id === id);
if (!task) {
return res.status(404).json({
success: false,
error: "Task not found"
});
}
res.json({
success: true,
data: task
});
});
app.post("/api/tasks", authCheck, (req, res) => {
const { title, completed = false } = req.body;
if (!title || title.trim() === "") {
return res.status(400).json({
success: false,
error: "Title is required"
});
}

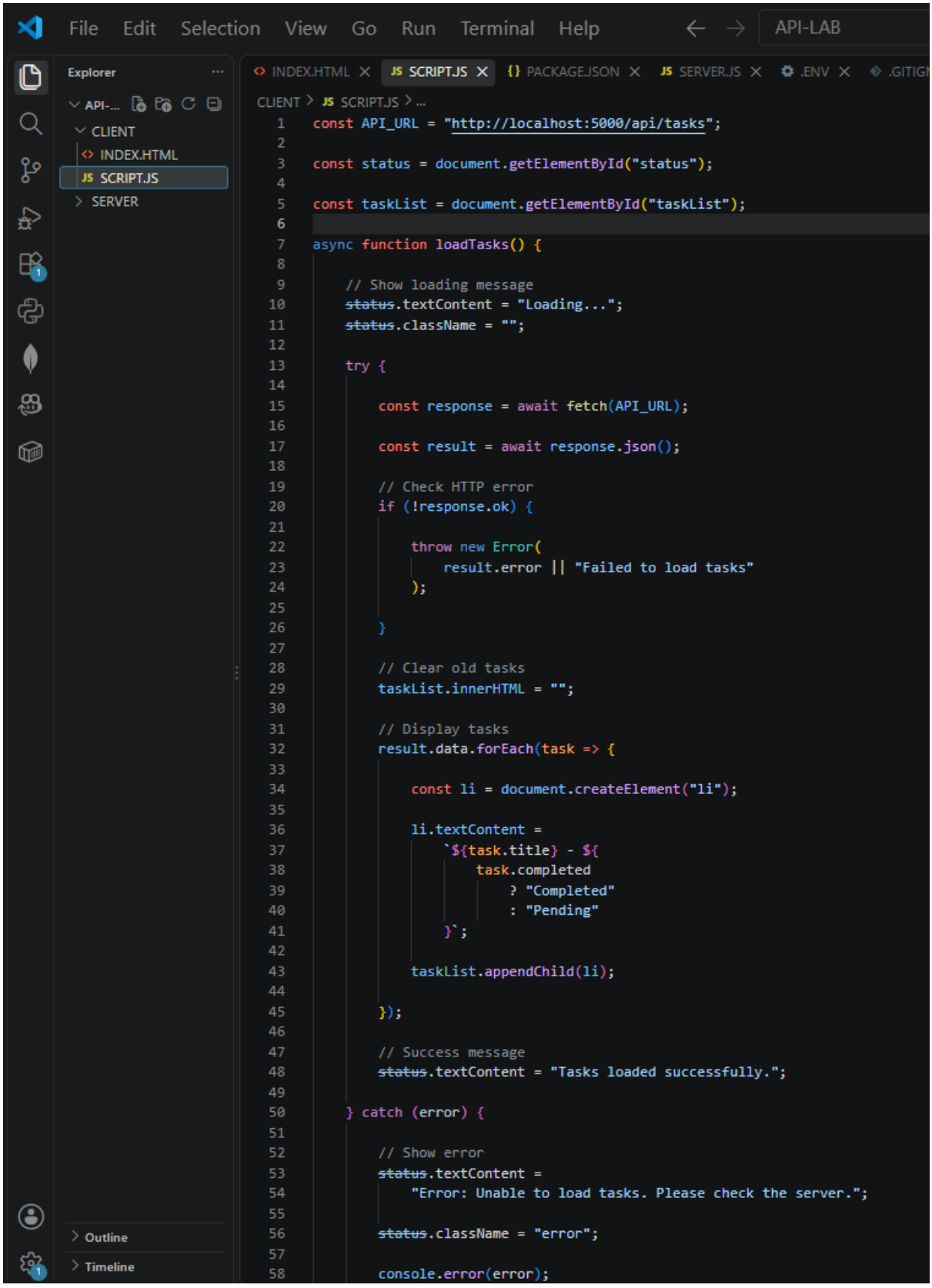

File Edit Selection View Go Run Terminal Help
API-LAB
Explorer
API-LAB
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.ENV
.GITIGNORE
package-lock.json
PACKAGE.JSON
SERVER.JS
Outline
Timeline
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
.ENV
.GITIGNORE
package-lock.json
SERVER > SERVER.JS > ...
104 app.post("/api/tasks", authCheck, (req, res) => {
111 success: false,
112 error: "Title is required"
113 });
114
115 }
116
117 const newTask = {
118 id: tasks.length > 0
119 ? Math.max(...tasks.map(task => task.id)) + 1
120 : 1,
121
122 title: title.trim(),
123
124 completed: completed
125 };
126
127 tasks.push(newTask);
128
129 res.status(201).json({
130 success: true,
131 data: newTask
132 });
133
134 });
135
136 app.put("/api/tasks/:id", authCheck, (req, res) => {
137 const id = Number(req.params.id);
138
139 const task = tasks.find(task => task.id === id);
140 if (!task) {
141
142 return res.status(404).json({
143 success: false,
144 error: "Task not found"
145 });
146
147 }
148
149 const { title, completed } = req.body;
150
151 if (title !== undefined) {
152 task.title = title;
153 }
154
155 if (completed !== undefined) {
156 task.completed = completed;
157 }
158
159 res.json({
160 success: true,
161 data: task
162 });
163
164 });
165
166 app.delete("/api/tasks/:id", authCheck, (req, res) => {
167
0 0

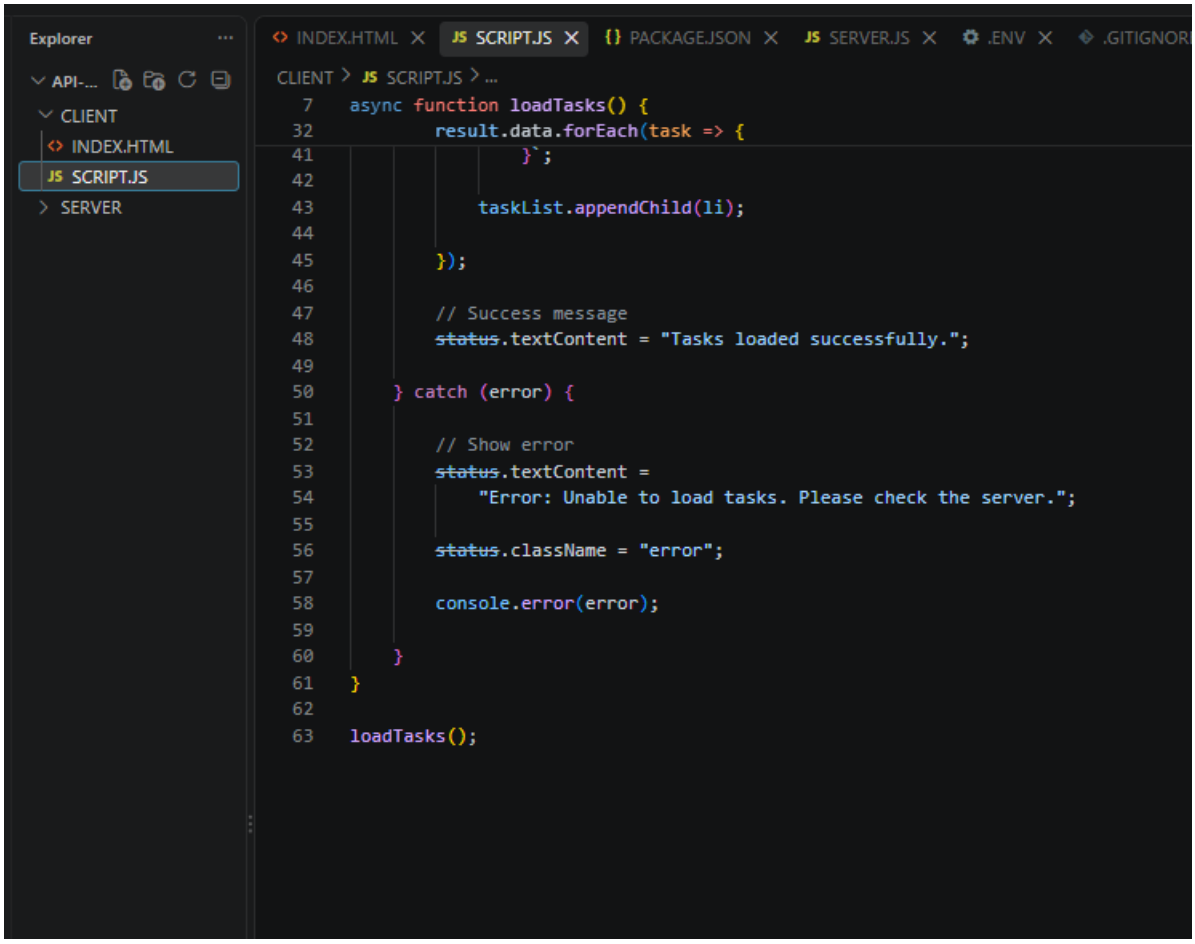

File Edit Selection View Go Run Terminal Help
API-LAB
Explorer
API-LAB
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.ENV
.GITIGNORE
package-lock.json
PACKAGE.JSON
SERVER.JS
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
.ENV
.GITIGNORE
SERVER > SERVER.JS > ...
164 });
165
166 app.delete("/api/tasks/:id", authCheck, (req, res) => {
167
168 const id = Number(req.params.id);
169
170 const taskIndex = tasks.findIndex(task => task.id === id);
171
172 if (taskIndex === -1) {
173
174 return res.status(404).json({
175 success: false,
176 error: "Task not found"
177 });
178
179 }
180
181 const deletedTask = tasks.splice(taskIndex, 1)[0];
182
183 res.json({
184 success: true,
185 message: "Task deleted successfully",
186 data: deletedTask
187 });
188 });
189
190 app.use((req, res) => {
191 res.status(404).json({
192 success: false,
193 error: "Route not found"
194 });
195 });
196
197 app.use((err, req, res, next) => {
198
199 console.error(err);
200
201 res.status(500).json({
202 success: false,
203 error: "Internal Server Error"
204 });
205
206 });
207
208 app.listen(PORT, () => {
209
210 console.log(`Server running on http://localhost:${PORT}`);
211
212 });

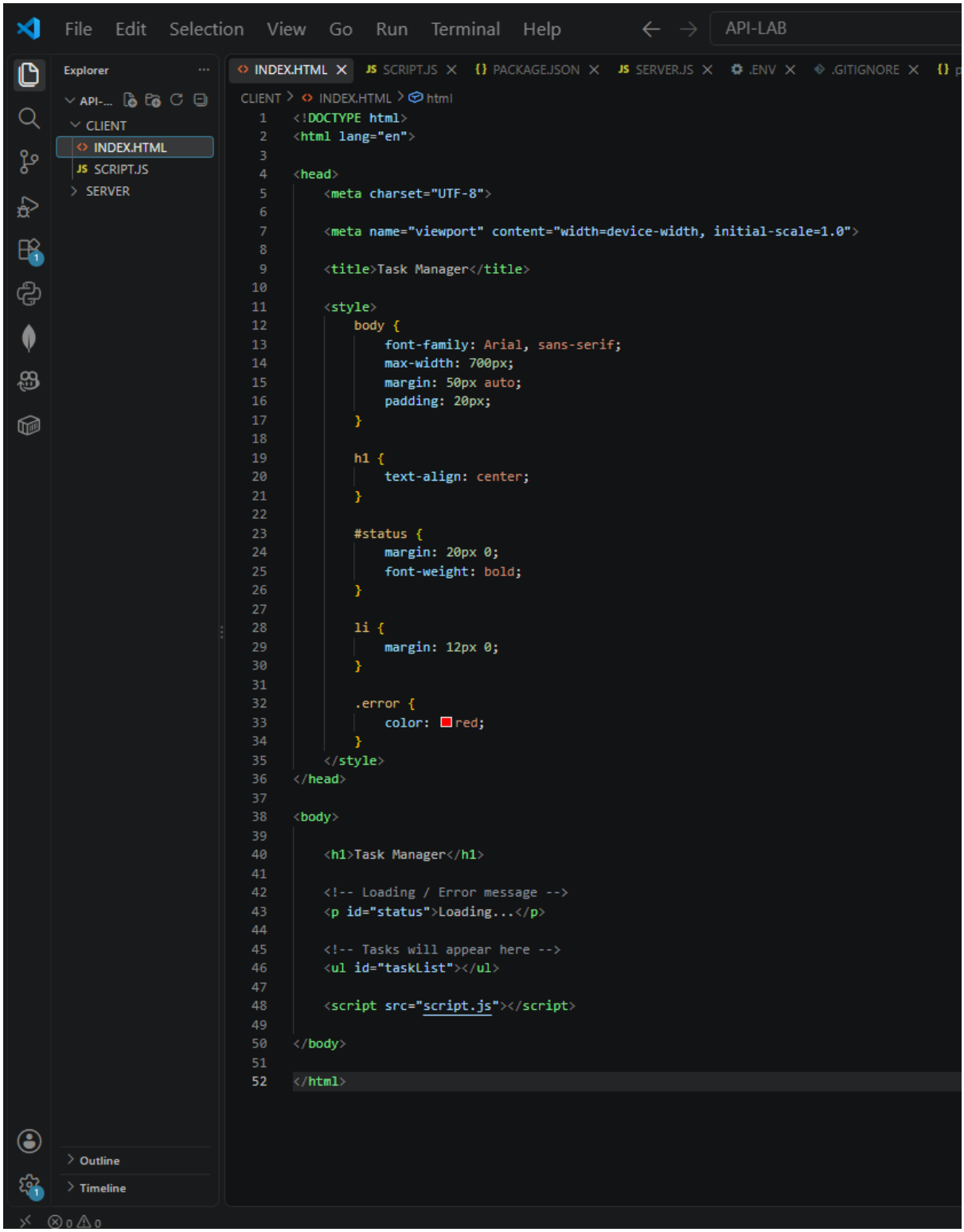

File Edit Selection View Go Run Terminal Help
Explorer
API-...
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.ENV
.GITIGNORE
package-lock.json
PACKAGE.JSON
SERVER.JS
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
SERVER > PACKAGE.JSON > ...
{
"name": "server",
"version": "1.0.0",
"description": "",
"main": "server.js",
"scripts": {
"start": "node server.js",
"dev": "nodemon server.js"
},
"keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"cors": "^2.8.5",
"dotenv": "^17.2.2",
"express": "^5.1.0"
},
"devDependencies": {
"nodemon": "^3.1.14"
}
}


File Edit Selection View Go Run Terminal Help
Explorer
API-...
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.ENV
.GITIGNORE
package-lock.json
PACKAGE.JSON
SERVER.JS
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER > package-lock.json > ...
{
"name": "server",
"version": "1.0.0",
"description": "Task Manager REST API",
"main": "server.js",
"scripts": {
"start": "node server.js",
"dev": "nodemon server.js"
},
"keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"cors": "^2.8.5",
"dotenv": "^17.2.2",
"express": "^5.1.0"
},
"devDependencies": {
"nodemon": "^3.1.10"
}
}


File Edit Selection View Go Run Terminal Help
Explorer
API-...
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
node_modules
.bin
accepts
anymatch
balanced-match
binary-extensions
body-parser
brace-expansion
braces
bytes
call-bind-apply-h...
call-bound
chokidar
content-disposition
content-type
cookie
cookie-signature
cors
debug
depd
dotenv
dunder-proto
ee-first
encodeurl
es-define-property
es-errors
es-object-atoms
escape-html
etag
express
fill-range
finalhandler
forwarded
fresh
function-bind
get-intrinsic
get-proto
glob-parent
gopd
has-flag
has-symbols
hasown
http-errors
Outline
Timeline
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
SERVER > .ENV
1 PORT=5000
2 API_KEY=mysecret123
0 0


File Edit Selection View Go Run Terminal Help
API-LAB
Explorer
API-...
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
.ENV
CLIENT > SCRIPT.JS > ...
const API_URL = "http://localhost:5000/api/tasks";
const status = document.getElementById("status");
const taskList = document.getElementById("taskList");
async function loadTasks() {
// Show loading message
status.textContent = "Loading...";
status.className = "";
try {
const response = await fetch(API_URL);
const result = await response.json();
// Check HTTP error
if (!response.ok) {
throw new Error(
result.error || "Failed to load tasks"
);
}
// Clear old tasks
taskList.innerHTML = "";
// Display tasks
result.data.forEach(task => {
const li = document.createElement("li");
li.textContent =
`${task.title} - ${
task.completed
? "Completed"
: "Pending"
}`;
taskList.appendChild(li);
});
// Success message
status.textContent = "Tasks loaded successfully.";
} catch (error) {
// Show error
status.textContent =
"Error: Unable to load tasks. Please check the server.";
status.className = "error";
console.error(error);
Outline
Timeline


Explorer
API-...
CLIENT
INDEX.HTML
SCRIPT.JS
SERVER
INDEX.HTML
SCRIPT.JS
PACKAGE.JSON
SERVER.JS
.ENV
CLIENT > SCRIPT.JS > ...
async function loadTasks() {
result.data.forEach(task => {
}`;
taskList.appendChild(li);
});
// Success message
status.textContent = "Tasks loaded successfully.";
} catch (error) {
// Show error
status.textContent =
"Error: Unable to load tasks. Please check the server.";
status.className = "error";
console.error(error);
}
}
loadTasks();




```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Task Manager</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            max-width: 700px;
            margin: 50px auto;
            padding: 20px;
        }

        h1 {
            text-align: center;
        }

        #status {
            margin: 20px 0;
            font-weight: bold;
        }

        li {
            margin: 12px 0;
        }

        .error {
            color: red;
        }
    </style>
</head>

<body>

    <h1>Task Manager</h1>

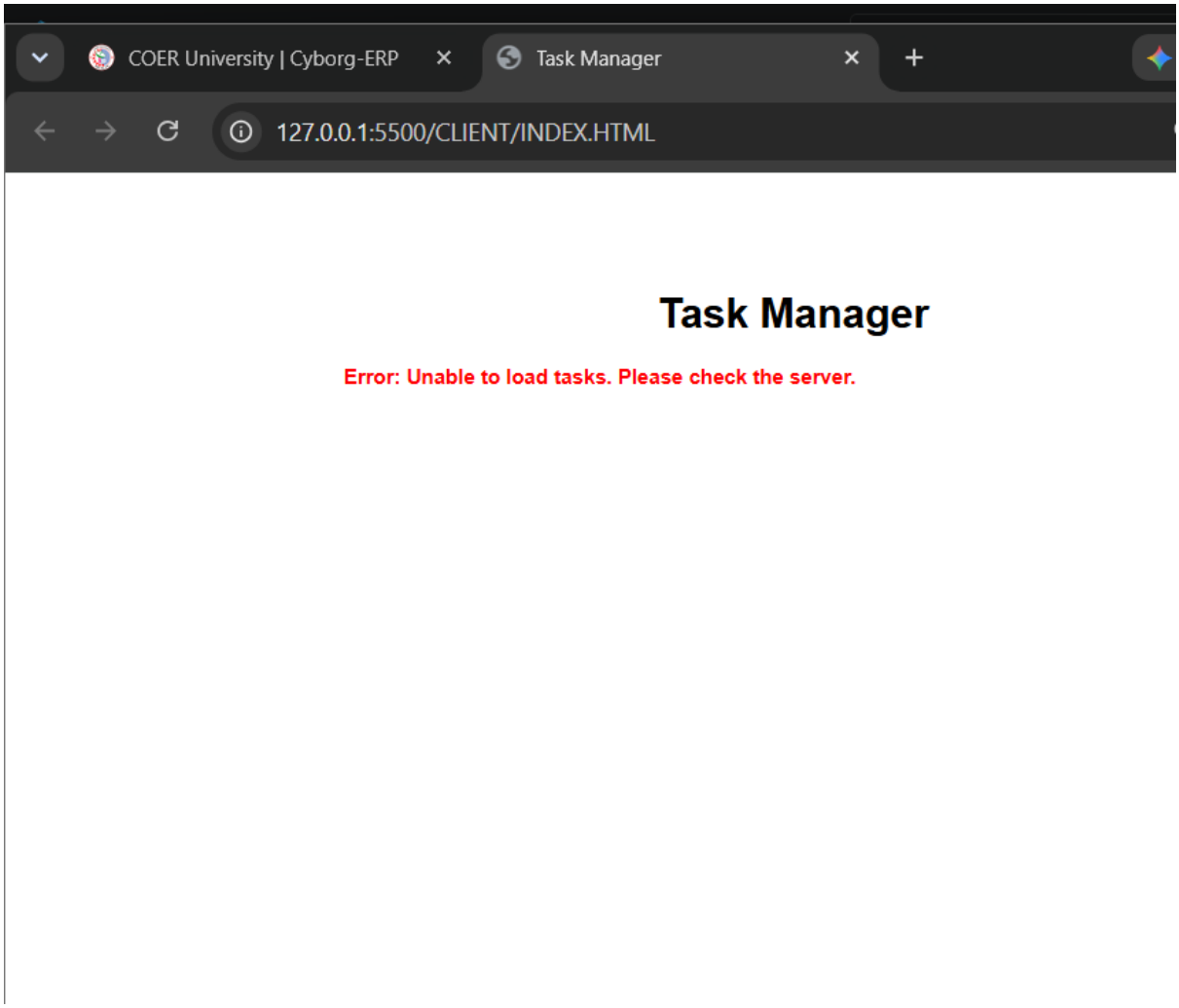
    <!-- Loading / Error message -->
    <p id="status">Loading...</p>

    <!-- Tasks will appear here -->
    <ul id="taskList"></ul>

    <script src="script.js"></script>

</body>

</html>
```


COER University | Cyborg-ERP
Task Manager
127.0.0.1:5500/CLIENT/INDEX.HTML
Task Manager
Error: Unable to load tasks. Please check the server.